

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	2 July 2026
Team ID	
Project Name	Library Book Request System using ServiceNow
Maximum Marks	4 Marks

S.No	Component	Description	Technology
1	User Interface	User interacts with the application through forms and lists.	ServiceNow Web UI
2	Application Logic-1	Validates that the issue date is not in the past.	ServiceNow Business Rule (JavaScript)
3	Application Logic-2	Automatically generates a unique request number.	ServiceNow Business Rule (JavaScript)

- 4 Application Logic-3 Sends email notification to the librarian after ServiceNow Notification

Technology Stack (Architecture & Stack)

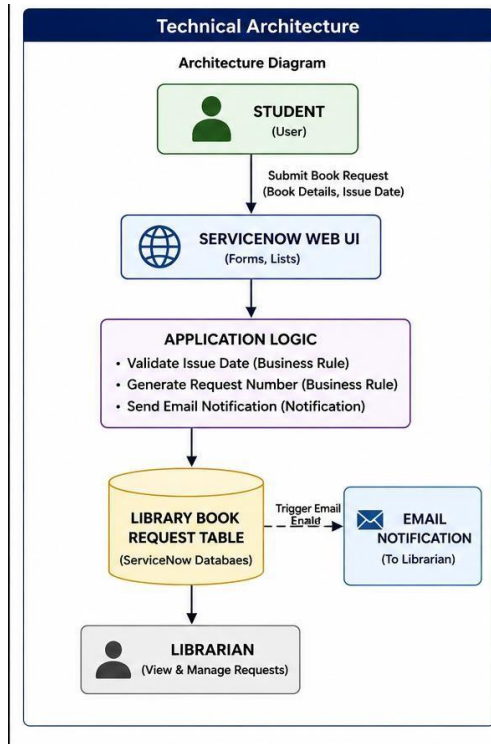


Table-1: Components & Technologies

S.No	Component	Description request submission.	Technology
5	Database	Stores library book request records.	ServiceNow Tables
6	Cloud Database	Stores application data in the ServiceNow cloud platform.	ServiceNow Cloud Database
7	File Storage	Stores application records and attachments.	ServiceNow Attachment Storage
8	External API-1	Not Used	N/A
9	External API-2	Not Used	N/A
10	Machine Learning Model	Not Used	N/A
Infrastructure (Server / Application hosted on the ServiceNow cloud			
11	ServiceNow Cloud	Cloud) platform.	

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source framework is used.	ServiceNow Platform
2	Security Implementations	Role-based access control and authenticated user access.	ServiceNow ACL & User Authentication

Cloud-based architecture supports increasing

3 Scalable Architecture ServiceNow Cloud Platform
numbers of requests.

Application is available online through the

4 Availability ServiceNow SaaS
ServiceNow platform.

Business Rules and Notifications provide

5 Performance ServiceNow Platform
efficient request processing.